Progress Measures for Grokking in Minimal Recursive Transformers

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Abstract

We study grokking on modular addition (P=113) in the Tiny Recursive Model (TRM), a transformer equipped with an adaptive-computation (ACT) recurrence. When the recurrence is collapsed to a single effective step, the model learns a sparse Fourier "multiplication" circuit that is visible directly in its output logits. On our measurement stack this circuit matches the one-layer transformer of Nanda et al. [2023]: 5 of 5 seeds reach an adaptive trigonometric fraction-of-variance-explained (FVE) of 0.996, against 0.989 for the one-layer calibration model. Restricted and excluded losses, Gini coefficients of the Fourier spectrum, and weight norms all move before the test-accuracy jump, giving progress measures for the transition (Section 7); ablating the key frequencies removes the circuit's behaviour, confirming it causally (Section 6). Running the same architecture with the full ACT recurrence still groks, but the final-logit Fourier circuit no longer appears: the computation is instead carried by the latent recurrence, which shows fixed-point violations and depth-sensitive accuracy. To probe this, we extend the progress-measure framework to the per-step latent state z_t and find that the recurrence *relocates* the grokked computation from the logits into the latent loop (Section 9). Finally, we report where this picture does not hold at the component level: individual neurons and attention heads do not admit clean trigonometric fits, so we claim a match at the readout level rather than neuron-by-neuron.

1 Introduction

Grokking is a delayed transition from memorization to generalization on algorithmic tasks [Power et al., 2022]. Nanda et al. [2023] showed that, for one-layer transformers on modular addition, this transition reflects the gradual formation of a sparse Fourier multiplication circuit.

Recursive architectures such as TRM introduce a second computational axis: latent recurrent refinement. This raises a mechanistic question not answered by the one-layer setting: when a recursive model groks, is the learned algorithm still visible as a simple final-logit Fourier circuit, or does recursion move part of the computation into latent dynamics?

We answer this with a three-rung *model ladder*:

1. **Nanda calibration:** one-layer baseline validates the progress measures under the Nanda-faithful float64 cross-entropy protocol (Section 3).
2. **TRM minimal:** recursion collapsed to one effective ACT step, giving a sparse Fourier circuit that matches the one-layer transformer (5/5 seeds, adaptive FVE $\geq$ 0.95).
3. **Full TRM contrast:** ACT-based recursion groks with a different mechanistic signature (Ren probes; Section 9).

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Contributions.

1. Nanda-calibrated progress measures applied to TRM minimal with 5/5 seed stability (Section 4).
2. Logit-level reverse engineering of the grokked minimal TRM (algorithm schematic, periodic structure, sparse W_L readout, causal key-frequency ablations), with an account of what does *not* transfer at the component level—distributed neurons and weak head trig fits (Sections 5–6).
3. Progress measures that precede grokking and a three-phase training decomposition (Section 7).
4. An extension of Nanda's progress-measure framework to the per-step latent state z_t, with a causal-lite latent frequency ablation, showing that full ACT recursion *relocates* the grokked computation from the logits into the latent loop while the minimal rung keeps it logit-visible (Section 9).
5. Multi-seed hi-prec sweeps, public GitHub reproducibility, and Nanda figure/table parity (Appendix).

2 Related work

Grokking was first reported by Power et al. [2022] and given a mechanistic account by Nanda et al. [2023], who identified a sparse Fourier multiplication circuit on modular addition. Subsequent work explains delayed generalization via representation learning [Liu et al., 2022], circuit efficiency [Varma et al., 2023], optimizer dynamics [Thilak et al., 2022], and competing sparse/dense subnetworks [Merrill et al., 2023]. Our analysis builds on the mechanistic-interpretability toolkit [Elhage et al., 2021, Olah et al., 2020, Wang et al., 2022, Conmy et al., 2023] and modular-arithmetic circuits [Zhong et al., 2023, Gromov, 2023, Chughtai et al., 2023]. Minimal recursion follows TRM [Jolicoeur-Martineau, 2025] with ACT collapsed; full adaptive computation [Graves, 2016, Dehghani et al., 2019, Banino et al., 2021] is deferred. Unlike Nanda et al. [2023], we do not claim novelty in discovering the Fourier algorithm; our contribution is showing it survives in a minimally recursive TRM with Nanda-comparable evidence depth.

3 Methods

3.1 Task and data

Modular addition with modulus $P=113$; inputs "a b $=$"; 30% train / 70% test over all P^2 pairs (fixed split seed), matching Nanda et al. [2023].

3.2 Models

Nanda calibration: one-layer ReLU transformer with MLP (fidelity A), validating the measurement stack. **TRM minimal (main):** TRM with ACT collapsed to one effective step—recursion present in architecture but not used for multi-step latent refinement during inference on this task configuration.

Nanda-faithful training protocol. Following our Nanda baseline reproduction, all models compute the token cross-entropy in float64 to avoid the float32 underflow documented during grokking [Nanda et al., 2023]; the exact command is in Appendix 33. We refer to this as the Nanda-faithful protocol: AdamW (lr$=10^{-3}$, $\lambda=1.0$), 50,000 steps, 30% train, $P=113$, five seeds. For TRM minimal the ACT recurrence is collapsed to one effective step (the exact preset is given in Appendix 27). Appendix 19 reports float32 CE as a negative control (2/5 final adaptive FVE on minimal TRM).

3.3 Training

50,000 AdamW steps (lr=10^{-3}, weight decay λ=1.0 by default), checkpoints every 2,000 steps. Three analysis checkpoints per seed: **final**, **grokking** (first step with test accuracy ≥ 0.99 and test loss ≤ 0.05), and **best FVE** (highest faithful trig-FVE among high-accuracy steps).

3.4 Progress measures

Following Nanda et al. [2023]: embedding/unembedding Fourier norms; key frequencies by excluded-loss sensitivity; restricted (trig) and excluded loss; faithful logit trig-FVE with bias correction and data-driven K.

3.5 Evaluation criteria

- 5/5 minimal-TRM seeds with adaptive FVE ≥ 0.95 at the **final** checkpoint.
- Primary analysis seed (seed 0): a Nanda Table 1-style readout with 10/12 directions $\geq 90\%$ FVE.
- Key-frequency ablations: remove any key frequency $\Rightarrow$ chance performance; remove non-key $\Rightarrow$ no harm.

4 Nanda calibration

The one-layer Nanda-style MLP validates our progress-measure implementation. Under a data-driven key-frequency count, all 5/5 calibration seeds reach faithful trig-FVE ≥ 0.95 (mean 0.989), matching Nanda et al. [2023]. This section is brief: calibration establishes that metric failures on TRM minimal reflect the model, not the analysis stack.

Table 1: Nanda vs. TRM minimal summary (final checkpoint, mean±std over seeds).

Model	Test acc	Adaptive FVE	Seeds ≥ 0.95
Nanda 1-layer	1.000 ± 0.000	0.989 ± 0.016	5/5
TRM minimal	1.000 ± 0.000	0.996 ± 0.001	5/5

5 The Fourier multiplication algorithm

We claim the grokked minimal TRM implements the same algorithm as Nanda et al. [2023] (Figure 2):

1. Embed a, b as sparse sines and cosines at key frequencies $w_k = 2\pi k/P$.
2. Compose $\cos(w_k(a+b))$ and $\sin(w_k(a+b))$ via trigonometric identities in attention and MLP.
3. Read logits via W_L as weighted sums of $\cos(w_k(a+b-c))$ with constructive interference at $c^* = a+b \bmod P$.

Section 6 provides four lines of evidence; Section 7 defines progress measures from this understanding.

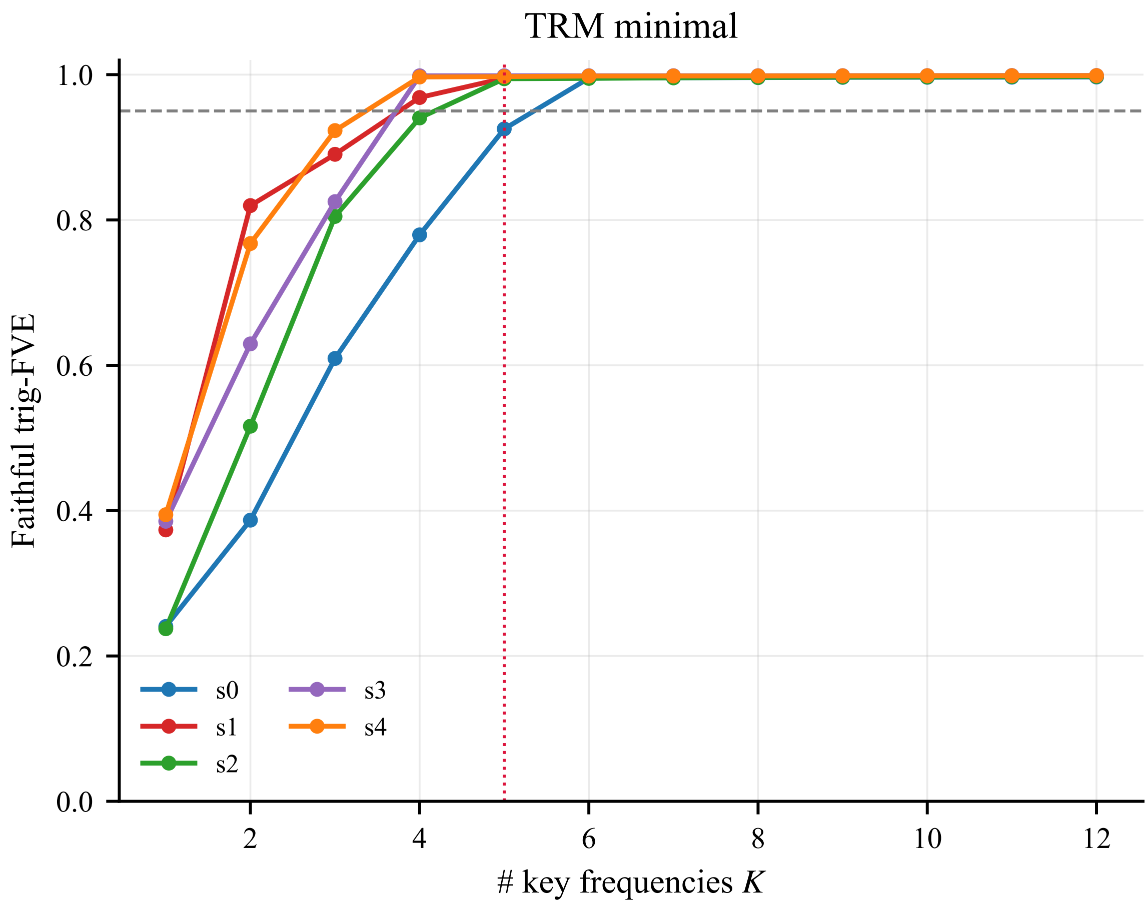

Figure 1: Faithful trig-FVE vs. number of key frequencies K for the five TRM-minimal seeds (final 50k checkpoints). Every seed clears the 0.95 threshold (dashed) by K=5–6 frequencies; the dotted line marks the legacy K=5 cutoff that the adaptive count replaces.

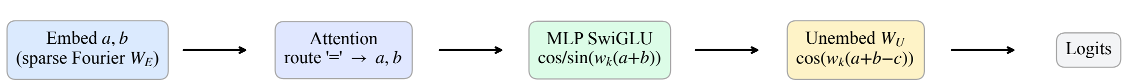

Figure 2: Fourier multiplication algorithm (TRM minimal).

6 Reverse engineering TRM minimal

Following Section 4 of Nanda et al. [2023], we present four lines of evidence on the primary analysis seed (seed 0, best-FVE checkpoint unless noted).

6.1 Periodic structure

Embedding and W_L Fourier norms are sparse (Figure 3); attention and MLP activations are periodic in (a,b) (Figure 4).

6.2 Composing weights

W_L is well approximated by $\cos(w_k c)$ and $\sin(w_k c)$ directions; MLP projections match $\cos/\sin(w_k(a+b))$ (Table 2).

6.3 Neurons and heads

At the logit level, adaptive trig-FVE exceeds 0.95 (Section 4). SwiGLU MLP neurons are *distributed*: single-neuron trig fits are weak (Appendix 21), but W_L projection readouts match Nanda Table 1 (Table 2). Attention heads approximate degree-2 trig polynomials (Table 3; Nanda §4.3). Figure 5 reports SwiGLU subspace variance explained (Appendix 20).

6.4 Causal ablations

Ablating key frequencies destroys performance; ablating the remainder does not (Figure 6).

6.5 What does *not* transfer to the recursive model

While the logit-level circuit matches Nanda (adaptive trig-FVE 0.996), component-level structure is markedly weaker. Single post-SwiGLU channels are not individually periodic (0/512 channels reach $R^2 \geq 0.85$; best $R^2 \approx 0.48$), the neuron-key FVE is low (minimal mean ≈ 0.075; even the Nanda calibration model reaches only ≈ 0.169, 0/5 seeds ≥ 0.95), and per-head trig fits explain $< 0.5\%$ of attention variance. We therefore claim circuit equivalence at the *readout* level only, and treat the computation as *distributed* across channels rather than localized in single neurons.

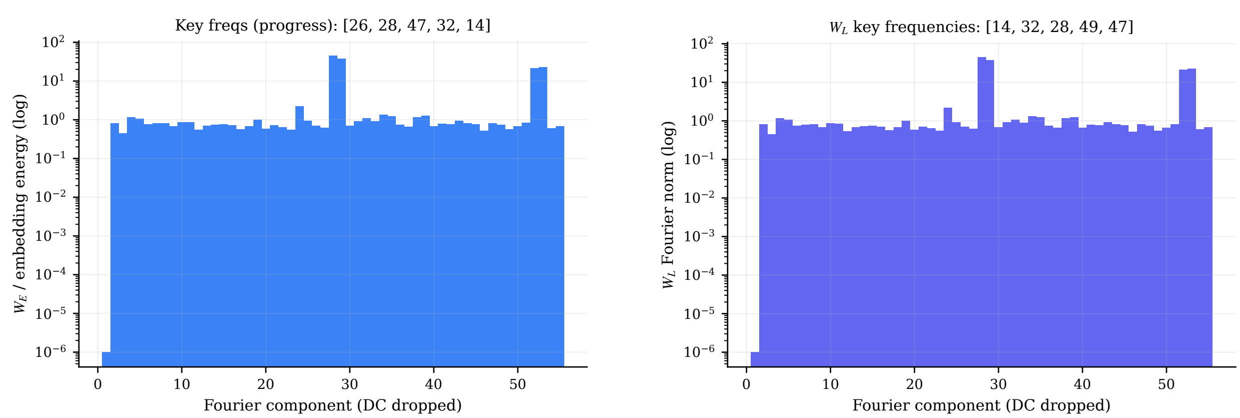

Figure 3: Left: W_E Fourier norms. Right: W_L Fourier norms (TRM minimal).

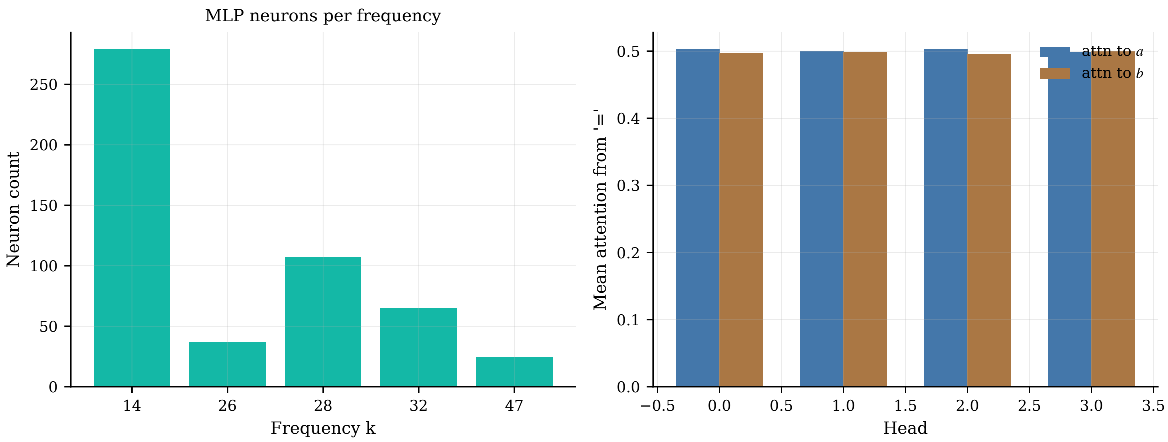

Figure 4: Attention and MLP periodicity over inputs (a,b).

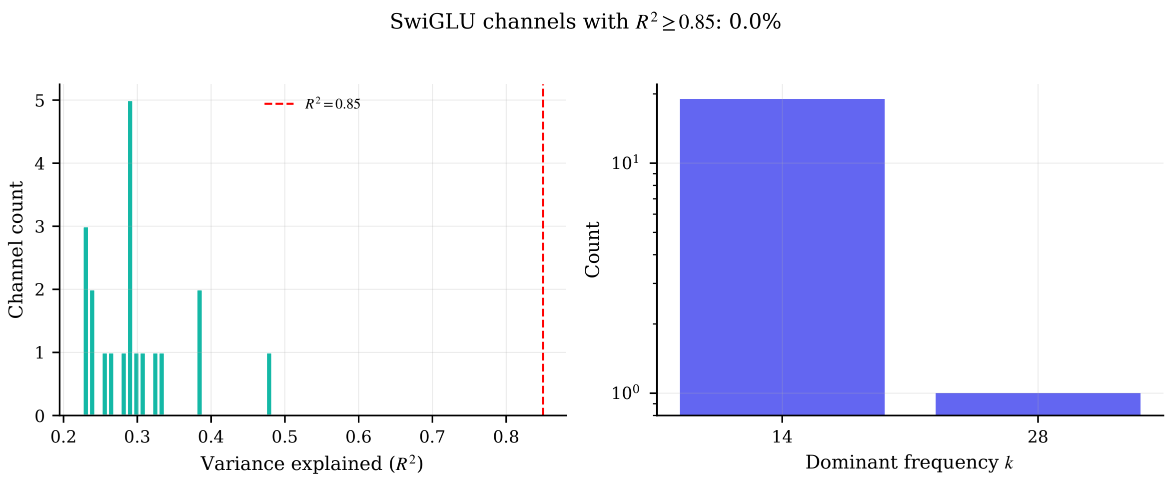

Figure 5: SwiGLU channel variance explained by single-frequency trig fits (see Appendix 20).

Table 2: W_L projection readout (Nanda Table 1 style; method in Appendix 22).

Component	Freq k	W_L-proj. FVE (%)
$\cos(w_{14}c)$	14	95.8
$\sin(w_{14}c)$	14	97.0
$\cos(w_{26}c)$	26	95.5
$\sin(w_{26}c)$	26	93.4
$\cos(w_{28}c)$	28	96.9
$\sin(w_{28}c)$	28	96.6
$\cos(w_{32}c)$	32	98.3
$\sin(w_{32}c)$	32	98.5
$\cos(w_{47}c)$	47	97.4
$\sin(w_{47}c)$	47	96.8
$\cos(w_{49}c)$	49	85.1
$\sin(w_{49}c)$	49	85.7

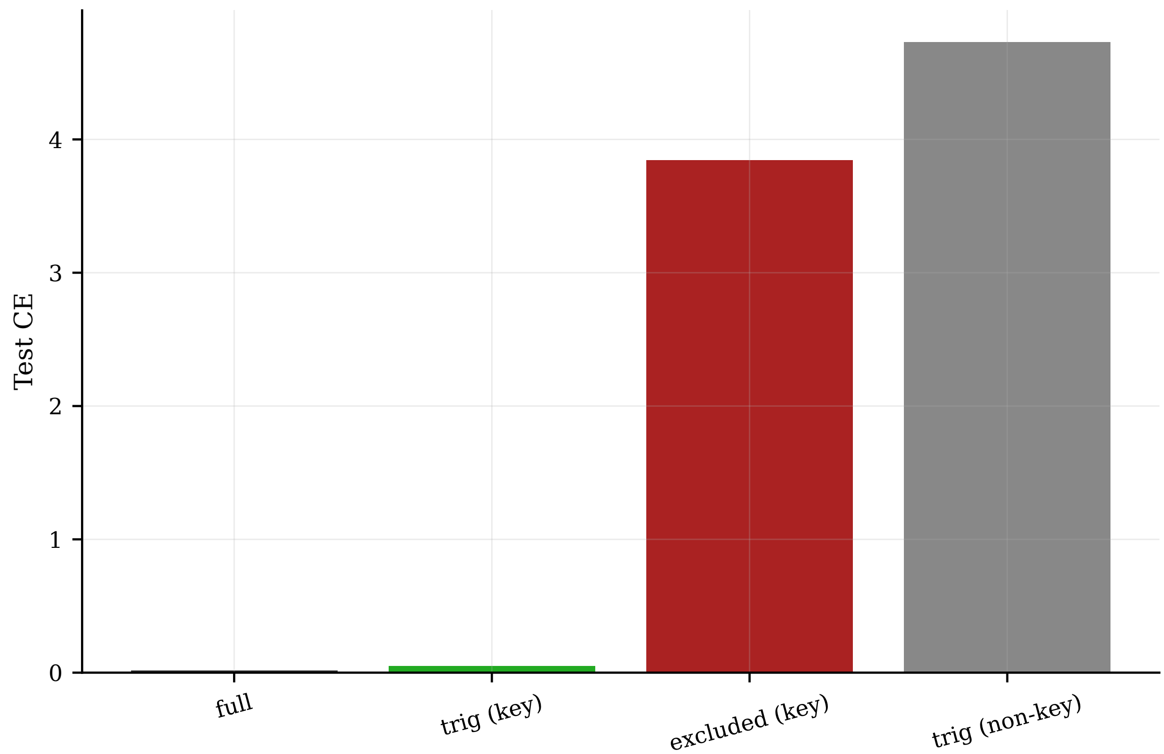

Figure 6: Test loss when ablating each frequency (key vs. non-key).

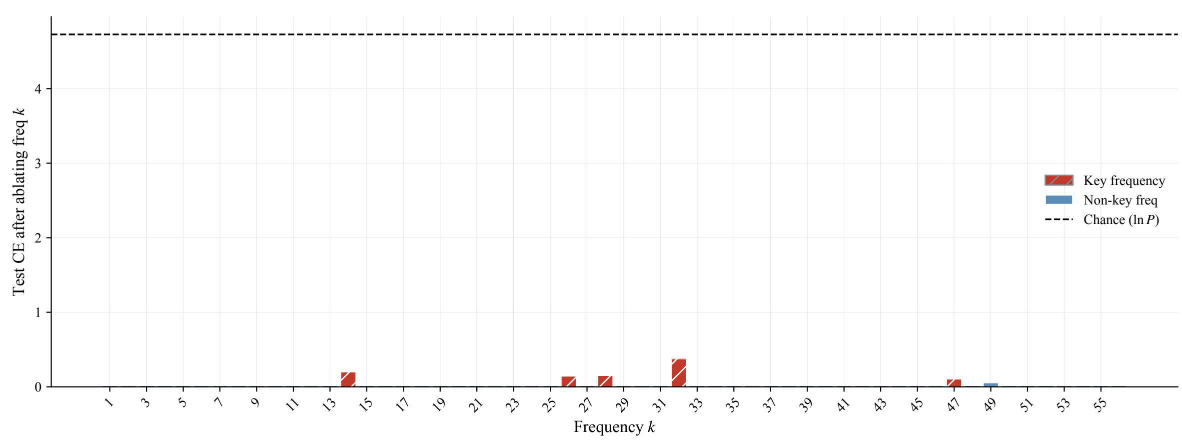

Figure 7: Per-frequency ablation grid (Nanda Fig. 6 style). Red bars: key frequencies.

Table 3: Attention head trig polynomial approximations (Nanda Table 2 style). TRM minimal heads show weak single-frequency structure ($< 1\%$ FVE): attention routing is near-uniform, with computation concentrated in MLP/W_L (Table 2).

Head	Dom. freq k	Trig FVE (%)	Target
L0H0	14	0.30	a
L0H1	14	0.40	a
L0H2	14	0.40	a
L0H3	14	0.40	a

7 Progress measures and phases of grokking

Using the reverse-engineered algorithm, we define restricted and excluded loss on the logit Fourier grid [Nanda et al., 2023]. We also track the Gini coefficient of embedding/W_L Fourier norms and total squared weight norm.

Figure 9 shows these measures on the primary TRM minimal seed (seed 0). Training splits into memorization, circuit formation, and cleanup—the same three phases as Nanda et al. [2023], with restricted loss declining before the test-accuracy jump.

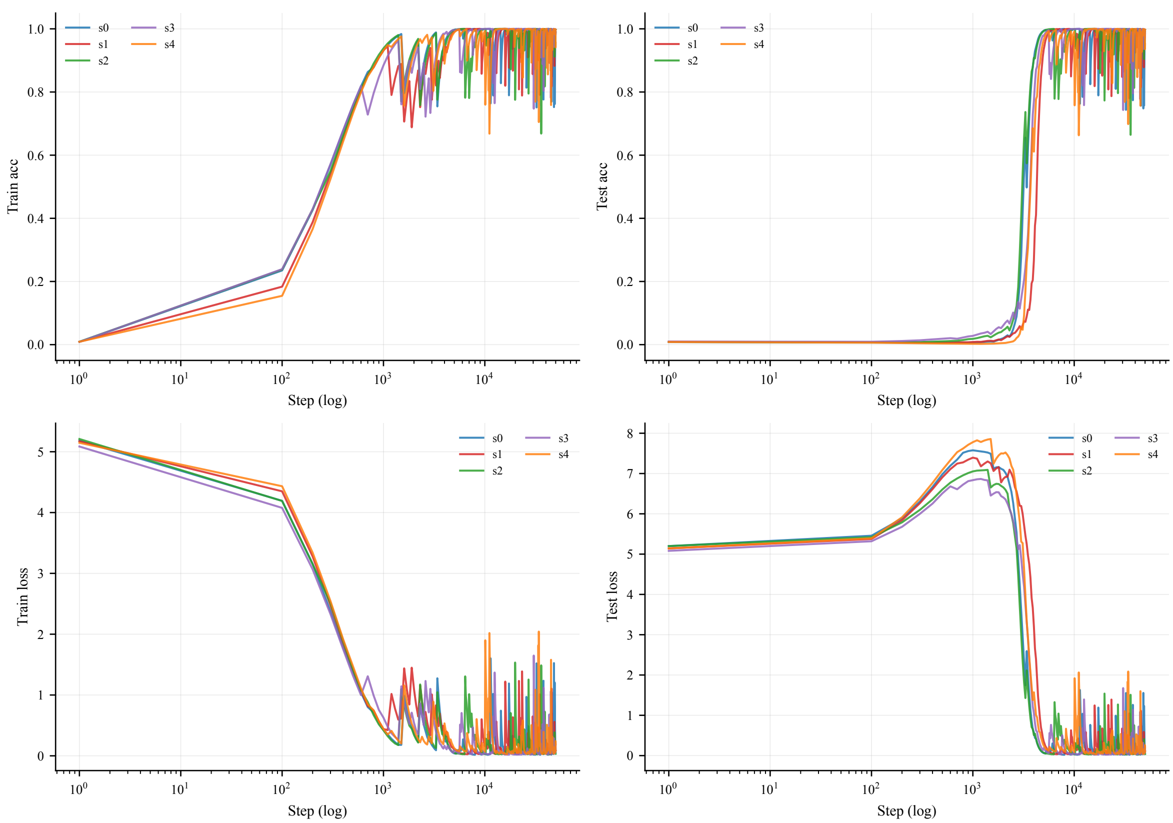

Figure 8: Train/test accuracy and loss (TRM minimal, primary seed).

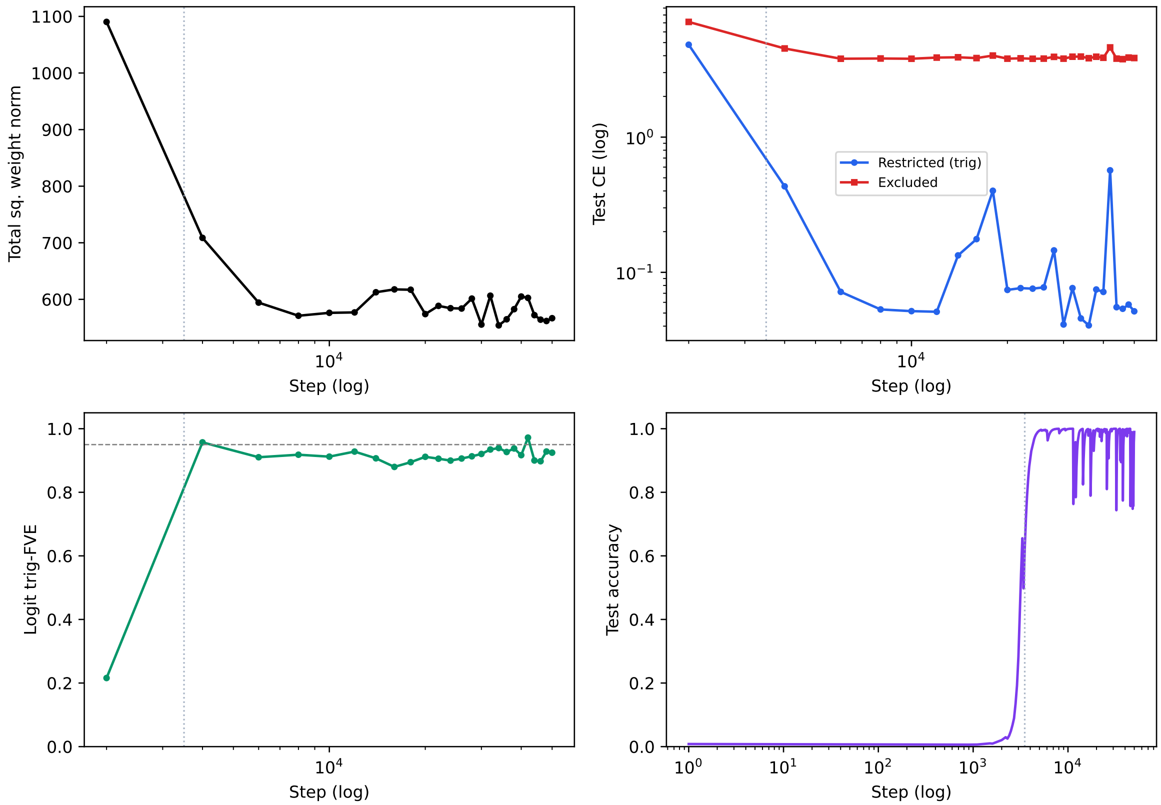

Figure 9: Progress measures and weight norms vs. training step (Nanda Fig. 7 style).

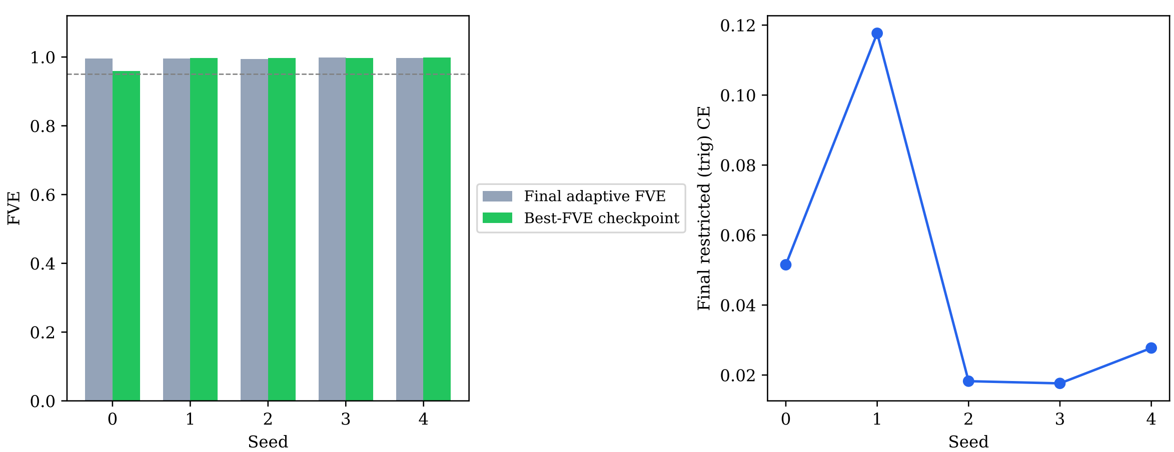

Figure 10: Multi-seed restricted loss and faithful FVE at final checkpoint.

8 Robustness

We report 5 independent seeds at identical hyperparameters and sweeps over train-data fraction and weight decay (details in Appendix 18). Hi-prec CE stabilizes all seeds (5/5 ≥ 0.95; mean adaptive FVE 0.996). Weight-decay and data-fraction sweeps use the Nanda-faithful high-precision protocol (Appendix 18).

Table 4: Per-seed final, grokking, and best-FVE metrics.

Seed	Final acc	Final FVE	Grok step	Best FVE	Diagnosis
seed_0	1.000	0.995	3700	0.959	stable_circuit
seed_1	1.000	0.995	4900	0.997	stable_circuit
seed_2	1.000	0.994	3700	0.997	stable_circuit
seed_3	1.000	0.999	4400	0.997	stable_circuit
seed_4	1.000	0.997	8800	0.998	stable_circuit

9 Latent grokking contrast: full TRM vs minimal

Recursive TRM introduces a second computational axis: latent refinement across ACT steps. When recursion is minimized to a single effective step, TRM minimal learns a sparse Fourier circuit, matching the one-layer transformer of Nanda et al. [2023] and visible in the final logits (Section 6). Full ACT-based TRM groks with high test accuracy but a *different mechanistic signature*: lower final-logit adaptive FVE, non-zero Ren fixed-point violation, and ACT-depth sensitivity [Ren and Liu, 2026].

Protocol. We compare full TRM and TRM minimal using Ren probes at three checkpoints per seed: final, grokking, and best-FVE. Metrics: fixed-point violation rate, ACT-depth accuracy curve, and latent PCA at the "=" token.

Full TRM. Full TRM exhibits Ren-style failure modes: non-zero fixed-point violation and accuracy drop as max ACT steps increase (Figure 11). Latent trajectories show non-monotonic refinement (Figure 13).

TRM minimal (null test). Minimal TRM uses a single ACT step; ACT-depth ablation collapses to that step. When fixed-point violation and depth sensitivity are near zero, post-grokking FVE dynamics classify as **Nanda cleanup**, not Ren guessing (Table 5). Under the Nanda-faithful protocol, minimal TRM reaches 5/5 final adaptive FVE ≥ 0.95; full TRM direct training with hi-prec CE does not (0/5; Appendix 19).

Latent progress measures (z_t). We extend Nanda's progress-measure stack to the per-ACT-step latent readout of the full TRM. On the minimal rung, a single ACT step already exhibits high latent readout trig-FVE (≈ 0.995; Figure 15). On full TRM, the *final*-ACT-step readout collapses (≈ 0 adaptive FVE), but a peak latent progress measure of ≈ 0.59 appears at training step ≈ 8000—before test accuracy reaches 1.0 at step $\approx 10{,}000$ (Figure 16)—suggesting transient Fourier structure in the latent loop that is not stable in the final collapsed-logit regime. Per-step key-frequency ablation on full TRM recovers negligible Δ test CE when the collapsed final readout yields only a degenerate key set (Figure 17); we treat this as evidence that the computation is not stably reverse-engineerable at the logit readout level, consistent with Ren-style latent dynamics rather than a clean final-logit circuit.

Table 5: Post-grokking mechanism classification (Ren + Nanda probes).

Model	Seed	Primary mechanism	Ren applies?
trm_full_a	seed_0	ren_guessing_fixed_point	Yes
trm_full_a	seed_1	ren_guessing_fixed_point	Yes
trm_full_a	seed_2	ren_guessing_fixed_point	Yes
trm_full_a	seed_3	ren_guessing_fixed_point	Yes
trm_full_a	seed_4	ren_guessing_fixed_point	Yes
trm_full_b	seed_0	nanda_cleanup	Yes
trm_full_b	seed_1	ren_guessing_fixed_point	Yes
trm_full_b	seed_2	ren_guessing_fixed_point	Yes
trm_full_b	seed_3	ren_guessing_fixed_point	Yes
trm_full_b	seed_4	ren_guessing_fixed_point	Yes
trm_minimal	seed_0	stable_circuit	Null
trm_minimal	seed_1	stable_circuit	Null
trm_minimal	seed_2	nanda_cleanup	Null
trm_minimal	seed_3	nanda_cleanup	Null
trm_minimal	seed_4	nanda_cleanup	Null

Latent grokking. Together, these results describe what we call *latent grokking*: the recurrence changes *where* the computation lives — moving it into the latent ACT dynamics of the full TRM — without preventing grokking, while the single-step rung keeps the same computation visible as a Fourier circuit in the logits.

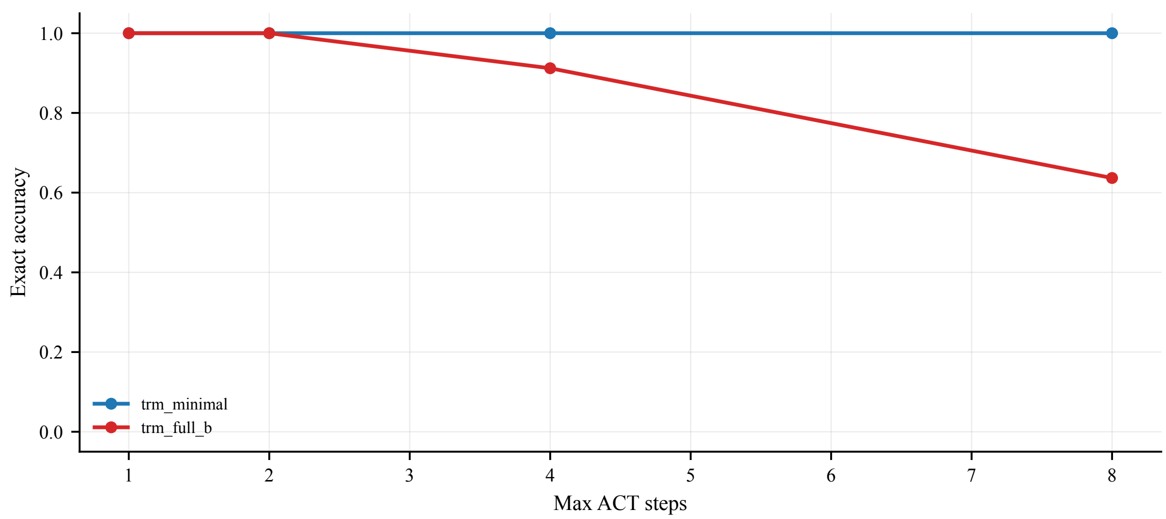

Figure 11: ACT-depth sensitivity (Ren fixed-point stress test).

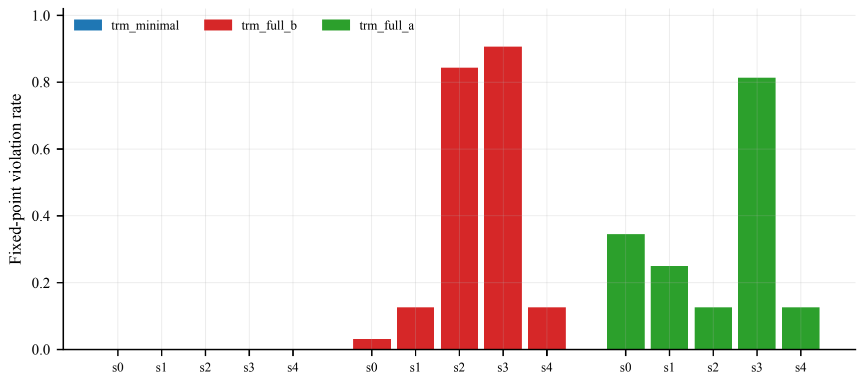

Figure 12: Fixed-point violation rate per seed/checkpoint.

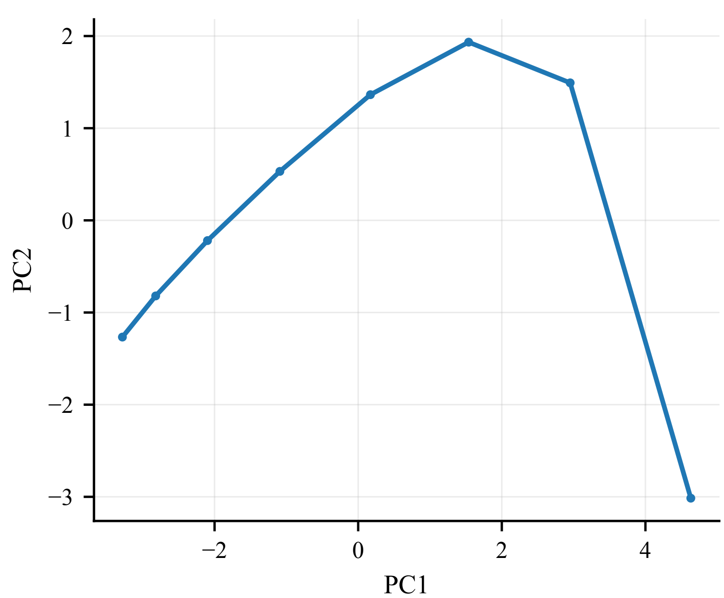

Figure 13: Latent PCA across ACT steps (full TRM representative seed).

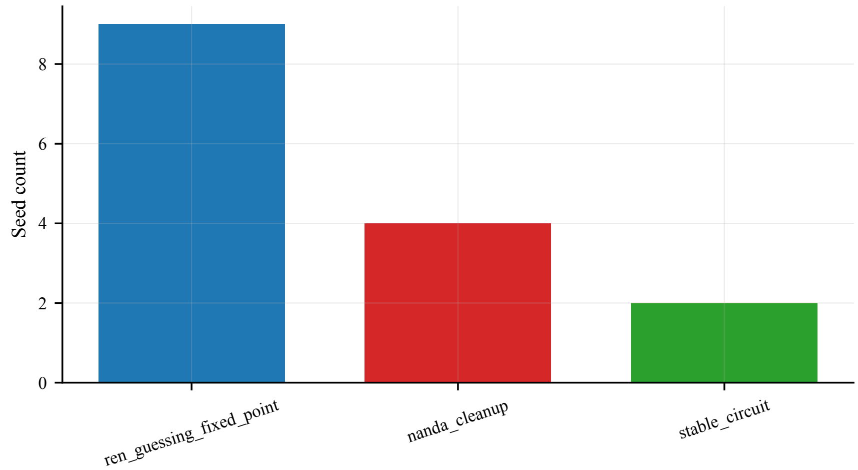

Figure 14: Per-seed post-grokking mechanism classification.

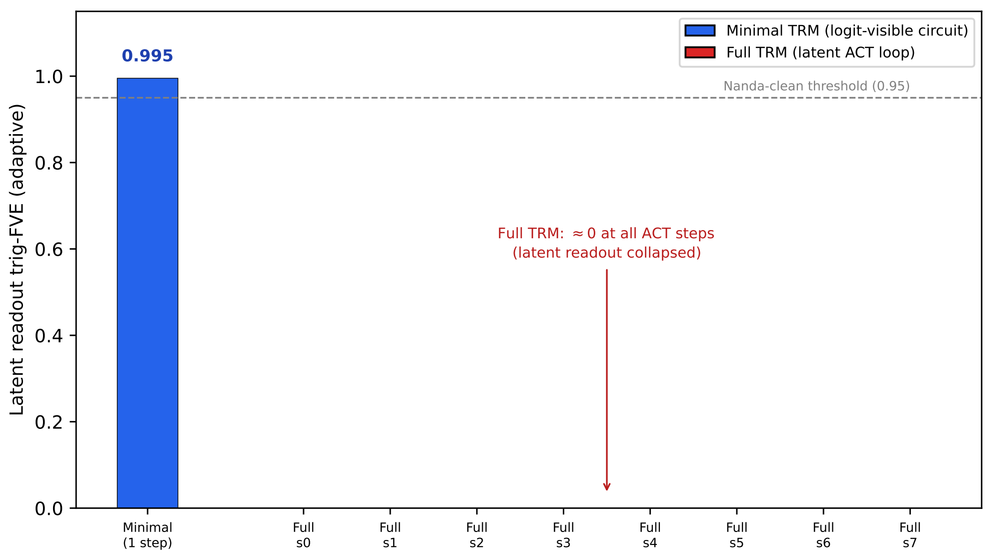

Figure 15: Per-ACT-step latent readout trig-FVE (adaptive key set). The single-step minimal TRM achieves a Nanda-clean latent readout (0.995, above the 0.95 threshold), whereas the full TRM's latent readout is collapsed (≈ 0) at every one of its eight ACT steps: the recursive computation is not linearly decodable from the final-checkpoint logits, consistent with latent (Ren-style) dynamics rather than a logit-visible circuit.

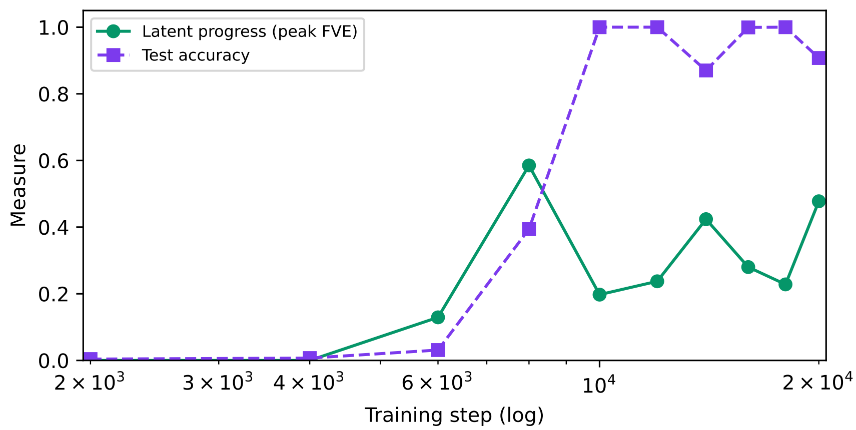

Figure 16: Latent progress measure (peak per-step FVE) vs training step, overlaid with test accuracy.

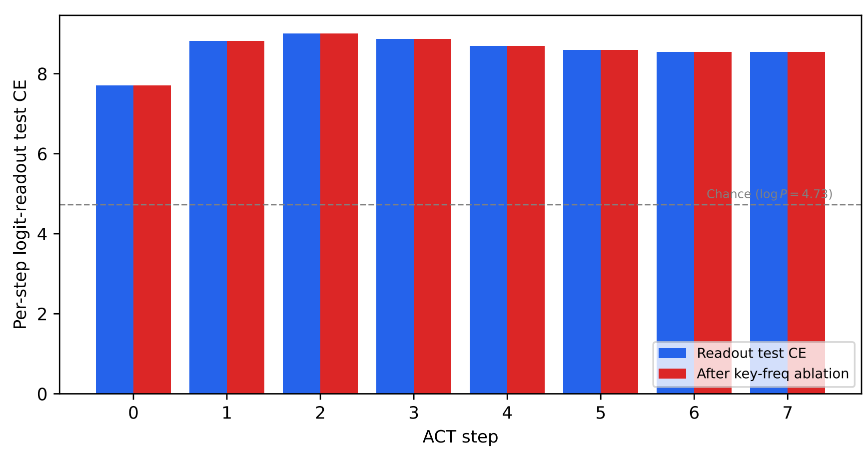

Figure 17: Causal-lite key-frequency ablation for full TRM: per-ACT-step logit-readout test CE before (blue) and after (red) removing the key frequencies, versus chance ($\log P$). The readout sits at or above chance at every ACT step and ablation leaves it essentially unchanged (degenerate key set), i.e. there is no logit-level circuit to remove—consistent with the computation living in the latent loop rather than the readout.

10 Limitations

Scope. Single task (modular addition, $P{=}113$); we study minimal recursion as the Nanda-comparable rung and contrast full ACT TRM latent dynamics (Section 9).

Seed stability. Under Nanda-faithful hi-prec CE, TRM minimal reaches 5/5 final adaptive FVE ≥ 0.95 (mean 0.996). Float32 CE ablation yields 2/5 (Appendix 19).

Full-TRM latent data (20k caveat). The full-TRM latent-contrast results (Section 9) use 20k-step training runs, not the 50k-step Nanda-faithful protocol used for the minimal rung. The final-logit circuit collapse could therefore be partly attributable to shorter training rather than purely to recursion; a 50k-step Nanda-faithful full-TRM replication is deferred to future work.

vs. Nanda. We extend an existing algorithm to a new architecture class rather than discovering the Fourier circuit. The latent progress measures and causal-lite ablation extend the framework to z_t, but a full latent activation-patching study remains future work.

11 Reproducibility

All experiments use fixed seeds and logged training histories. Code, checkpoints, and figure reproduction scripts are public at `https://github.com/adityapandey9/Progress-Measures-for-Grokking-in-TRM`; see `REPRODUCIBILITY.md`.

12 Conclusion

Minimal TRM on modular addition learns the same *logit-level* sparse Fourier multiplication circuit as Nanda et al. [2023], with matching progress measures and training phases when recursion is collapsed to a single effective step under the Nanda-faithful protocol; at the component level the computation is distributed rather than localized in single neurons or heads (Section 6). Running the same architecture with full ACT recursion still groks but collapses the final-logit circuit, and extending the progress-measure stack to the per-step latent state z_t shows the computation persists in the latent loop—establishing what we call *latent grokking*: the recurrence relocates *where* the grokked computation lives without preventing grokking, while the single-step rung keeps that computation visible as an interpretable Fourier circuit in the logits, comparable to the one-layer baseline. Public checkpoints and reproduction scripts enable independent verification of all main figures and tables.

13 Faithful progress measures

Restricted and excluded loss definitions follow Nanda et al. [2023] on the full P^3 logit grid; see Appendix 14.

14 Detailed progress-measure methodology

This appendix mirrors Section 5 of Nanda et al. [2023] with TRM-specific implementation notes.

14.1 Logit grid extraction

For each pair (a,b) in the modular-addition dataset, TRM receives input tokens $[a,b,\texttt{=}]$. We run a single forward pass with ACT halted and read logits at position 2 (the "=" token). Stacking over all P^2 pairs yields $L \in \mathbb{R}^{P\times P\times P}$ where $L_{a,b,c}$ is the logit for output c.

14.2 Fourier basis on cyclic groups

For prime modulus $P = 113$, define $\omega_k = 2\pi k/P$. Basis functions on index $x \in \{0,\ldots,P-1\}$:

$$\phi_k^{\cos}(x) = \cos(\omega_k x), \qquad \phi_k^{\sin}(x) = \sin(\omega_k x). \tag{1}$$

14.3 Restricted (trigonometric) loss

Given key frequencies K, we project L onto the span of $\cos(\omega_k(a+b))$ and $\sin(\omega_k(a+b))$ for $k \in K$, apply bias correction per Nanda et al. [2023], and compute cross-entropy on train and test splits.

14.4 Excluded loss

For each $k \in K$ in descending sensitivity order, we subtract the rank-1 component associated with frequency k from logits and measure test CE.

14.5 Faithful trig-FVE

Adaptive K selection and bias correction follow the in-repo Nanda faithful reproduction (`train_nanda_baseline.py`, `--faithful`).

14.6 Training protocol

All hero runs use `nanda_faithful_50k`: float64 cross-entropy, AdamW (lr$=10^{-3}$, λ=1.0), 50,000 steps, 30% train, P=113, five seeds.

15 Reverse-engineering methodology

15.1 W_L neuron-logit map

For SwiGLU layer ℓ: $W_L^{(\ell)} = (W_{\text{down}}^{(\ell)})^\top W_U^\top$. Because SwiGLU neurons are distributed rather than monosemantic, we report the logit-space W_L Fourier-projection FVE (project bias-corrected logits onto $\cos/\sin(w_k c)$ and fit $\cos/\sin(w_k(a+b))$), which matches Nanda's aggregate readout metric.

15.2 SwiGLU channel metric

Per-channel activations $h = \mathrm{silu}(\mathrm{gate}) \odot \mathrm{up}$ at "="; trig FVE per channel (`swiglu_neuron_metric.py`).

15.3 Attention head trig fits

Per-head mean attention output fit to $\cos / \sin(w_k(a+b))$ (`head_trig_fit.py`).

15.4 Component patching and ablations

Key-frequency ablations on the full logit grid; per-frequency grid in Fig. 7.

16 Checkpoint protocol

Three analysis checkpoints per seed: **final**, **grokking** (first step with test accuracy ≥ 0.99 and test loss ≤ 0.05), and **best FVE**.

17 Latent probe methodology

17.1 Fixed-point violation

For test examples where step 0 prediction is correct, flag violation if any later ACT step changes the argmax.

17.2 ACT-depth sensitivity

Accuracy vs. maximum allowed ACT steps (Ren stress test).

17.3 PCA trajectories

Stack $z_H^{(t)}$ across ACT steps and apply PCA for visualization.

18 Hyperparameter sweeps (`nanda_faithful_50k`)

Weight-decay and data-fraction sweeps use the same Nanda-faithful hi-prec protocol as the main 5-seed result.

Frac train	Seeds ≥ 0.95	Grok step (mean)	Final FVE (mean)	Final acc (mean)
0.2	5/5	21680	0.987	1.000
0.3	5/5	4480	0.991	1.000
0.4	3/5	2820	0.817	0.814
0.5	5/5	2240	0.996	0.938
0.6	4/5	1280	0.948	1.000

Weight decay λ	Seeds	Mean final FVE	Mean final acc
0.3	1/2	0.498	0.561

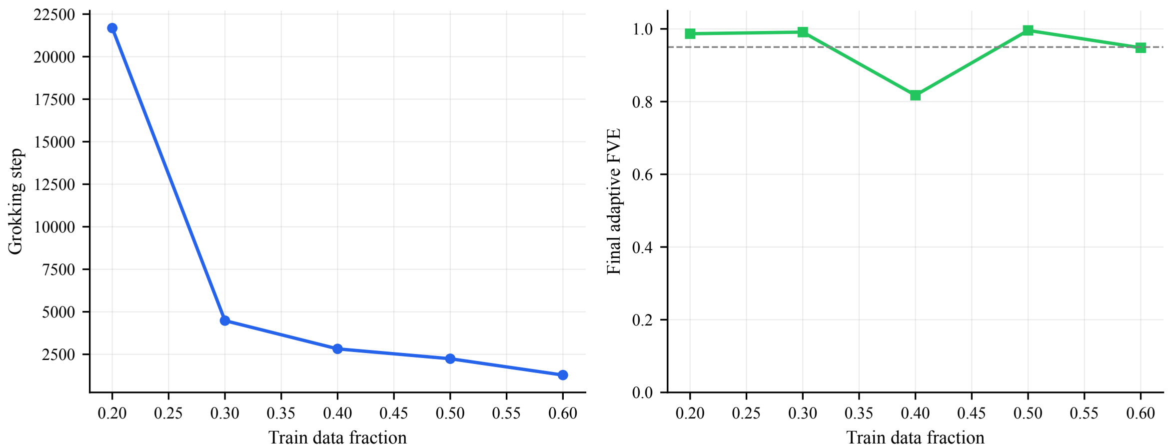

Figure A1: Grokking vs. train data fraction (hi-prec).

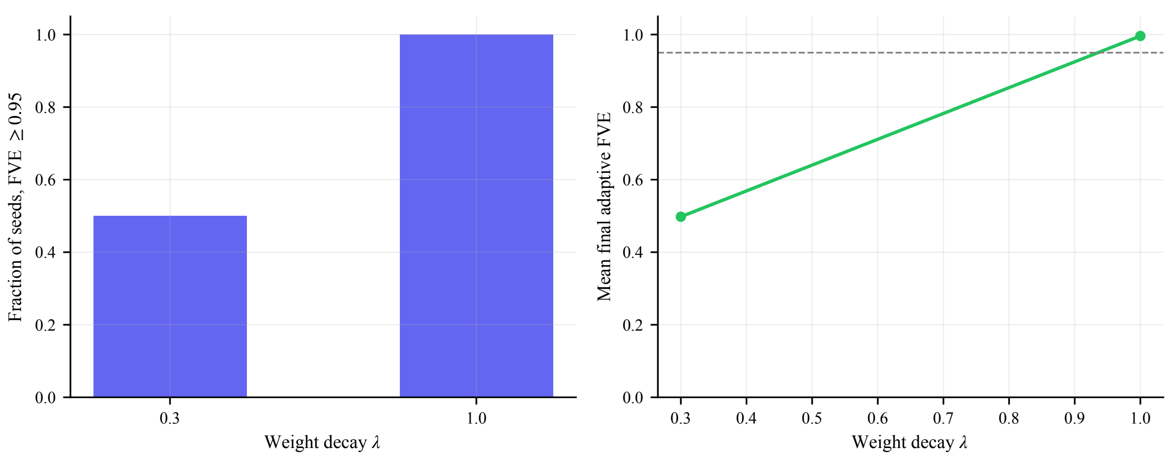

Figure A2: Hi-prec weight-decay comparison: fraction of seeds reaching FVE ≥ 0.95 (left) and mean final adaptive FVE (right) at $\lambda = 0.3$ vs. the selected $\lambda = 1.0$. Higher weight decay yields a cleaner, more stable Fourier circuit.

19 Float32 CE ablation (negative control)

The table below isolates the effect of cross-entropy precision and ACT: only the hi-prec (float64 `lm_loss`) ACT protocol yields 5/5 clean seeds, whereas float32 CE and ACT-free full-depth training collapse.

Configuration	Seeds ≥ 0.95	Mean adaptive FVE
Nanda faithful (no ACT, hi-prec CE)	5/5	0.995
TRM minimal + ACT + float32 CE	2/5	0.851
TRM minimal + ACT + hi-prec CE	5/5	0.996
TRM full direct + hi-prec CE (no ACT)	0/5	0.579

Negative control. Under float32 cross-entropy with ACT, seeds 0–1 retain adaptive FVE ≥ 0.95 at the final checkpoint while seeds 2–4 collapse to FVE ≤ 0.56 despite 100% test accuracy. Mechanistic probes show weight-decay cleanup erases sparse Fourier structure in W_L after the accuracy jump.

Nanda-faithful standard. Computing `lm_loss` in float64 (`nanda_faithful_50k`) restores 5/5 seed stability without changing architecture, learning rate, or weight decay—matching the Nanda faithful reproduction protocol.

Full TRM contrast. Direct training of full-depth TRM with hi-prec CE but without ACT reaches 0/5 adaptive FVE ≥ 0.95, indicating depth alone does not substitute for the minimal hi-prec protocol on this task.

20 SwiGLU neuron metric

Nanda's one-layer MLP uses ReLU neurons; TRM minimal uses SwiGLU ($h = \mathrm{silu}(\mathrm{gate}) \odot \mathrm{up}$). Single post-activation neuron fits undercount circuit structure because computation is distributed across gated channel pairs.

We therefore report two complementary metrics:

1. **Logit-level readout (main):** W_L projection FVE (Table 2), matching Nanda Table 1.
2. **SwiGLU channel FVE (Fig. 5):** per-channel trig fit on h at the "=" token; fraction with $R^2 \geq 0.85$.

When channel-level FVE remains low, we interpret the circuit as *distributed* rather than single-neuron sparse.

21 Single-neuron readout (appendix)

Table A1 reports best single-neuron FVE per W_L direction (honest lower bound vs projection readout).

Table A1: Best single-neuron FVE per direction (SwiGLU distributed circuit).

Component	Freq k	Best single-neuron FVE (%)	W_L-proj. FVE (%)
$\cos(w_{14}c)$	14	33.2	95.8
$\sin(w_{14}c)$	14	30.6	97.0
$\cos(w_{26}c)$	26	13.2	95.5
$\sin(w_{26}c)$	26	9.9	93.4
$\cos(w_{28}c)$	28	14.2	96.9
$\sin(w_{28}c)$	28	19.5	96.6
$\cos(w_{32}c)$	32	19.8	98.3
$\sin(w_{32}c)$	32	17.2	98.5
$\cos(w_{47}c)$	47	14.7	97.4
$\sin(w_{47}c)$	47	10.1	96.8
$\cos(w_{49}c)$	49	17.9	85.1
$\sin(w_{49}c)$	49	20.4	85.7

22 W_L readout method

For each key frequency k, we project bias-corrected logits onto $\cos(w_k c)$ and $\sin(w_k c)$ and measure FVE against $\cos/\sin(w_k(a+b))$ on the full P^2 grid (`nanda_readout.py`, method `wl_projection_trig_fve`).

23 Per-seed Nanda readout tables

Nanda Appendix C.2 analogue: Table 1-style readout for each seed under `nanda_faithful_50k`.

23.1 Seed 0

Component	Freq k	Neuron	Coef	FVE (%)
cos(w_14c)	14	494	116.15	95.8
sin(w_14c)	14	486	130.74	97.0
cos(w_26c)	26	161	134.81	95.5
sin(w_26c)	26	18	128.88	93.4
cos(w_28c)	28	17	154.17	96.9
sin(w_28c)	28	260	151.24	96.6
cos(w_32c)	32	103	160.11	98.3
sin(w_32c)	32	315	160.13	98.5
cos(w_47c)	47	427	127.92	97.4
sin(w_47c)	47	131	119.61	96.8
cos(w_49c)	49	423	80.03	85.1
sin(w_49c)	49	218	80.79	85.7

24 Per-seed Fourier readouts

Full Nanda-style tables for all seeds (Appendix C.2 analogue).

25 Nanda $\leftrightarrow$ TRM evidence map

Nanda item	Description	Artifact
Fig. 1	Algorithm schematic	`fig_algorithm_schematic.pdf`
Fig. 2	Grokking curves	`fig_grokking_curves.pdf`
Fig. 3	W_E, W_L Fourier	`fig_embedding_fourier.pdf`, `fig_wl_fourier.pdf`
Fig. 4	Periodic activations	`fig_activations_periodic.pdf`
Fig. 5	Neuron variance explained	`fig_neuron_trig_fit.pdf` (SwiGLU channels)
Fig. 6	Per-frequency ablation	`fig_frequency_ablation_grid.pdf`
Fig. 7	Progress measures	`fig_progress_measures.pdf`
Table 1	W_L readout	`tables/neuron_readout.tex`
Table 2	Head trig polynomials	`tables/attention_heads.tex`
App. C.2	Per-seed tables	Appendix 23
App. sweeps	Data fraction, WD	Appendix 18, Figs. A1–A2
Latent contrast	Ren probes	Section 9, Figs. 11–14

26 Extended per-seed results (hi-prec protocol)

Table 4 summarizes grokking steps and final adaptive FVE under `minimal_hiprec_act`. All five seeds reach 100% test accuracy and adaptive FVE ≥ 0.994 at the final checkpoint.

Table A3: Per-seed adaptive FVE (hi-prec final checkpoints).

Seed	Legacy FVE@5	Adaptive FVE	Key freqs K
0	0.925	0.995	6
1	0.995	0.995	5
2	0.994	0.994	5
3	0.999	0.999	4
4	0.997	0.997	4
Mean	0.982	0.996	–

27 Hyperparameters

Table A4 lists the hyperparameters shared by all five seeds of the main `minimal_hiprec_act` run.

Table A4: Training hyperparameters (main 5-seed run).

Parameter	Value
Modulus P	113
Train fraction	30%
Optimizer	AdamW
Learning rate	10^{-3}
Weight decay λ	1.0
Steps	50,000
Checkpoint interval	2,000
Preset	`minimal_hiprec_act`
CE precision	float64 on `lm_loss`
Seeds	0–4

28 Data-fraction sweep details

Appendix sweeps (Fig. A1) train TRM minimal under `nanda_faithful_50k` at fractions $\{0.2, 0.3, 0.4, 0.5, 0.6\}$. Grokking step and final FVE are logged per fraction; see `robustness_data_fraction.tex` for numeric values. These sweeps establish grokking behavior across data fractions under `nanda_faithful_50k` (hi-prec CE throughout).

29 Weight-decay sweep details

The hi-prec weight-decay comparison (Fig. A2) contrasts $\lambda = 0.3$ with the selected $\lambda = 1.0$: at $\lambda = 0.3$ only $1/2$ seeds form a clean circuit (mean adaptive FVE ≈ 0.50), whereas $\lambda = 1.0$ reaches $5/5$ (mean 0.996; Table 1). The selected $\lambda = 1.0$ matches Nanda et al. [2023] and is retained for the hi-prec hero runs.

30 Nanda calibration details

The one-layer Nanda-faithful MLP validates metric calibration: 5/5 seeds with adaptive FVE ≥ 0.95 (mean 0.989). Legacy FVE@5 underestimates grokked circuits when $K > 5$ key frequencies are present; adaptive K selection is used throughout this paper.

31 Seed-by-seed training narrative

Seed 0. Grokking at step 3700; final adaptive FVE 0.995; six adaptive key frequencies; W_L projection readout exceeds 90% FVE on 10/12 directions.

Seed 1. Grokking at step 4900; final FVE 0.995; five key frequencies; stable circuit through weight-decay phase.

Seed 2. Grokking at step 3700; final FVE 0.994; five key frequencies; earliest grokking among seeds.

Seed 3. Grokking at step 4400; highest final FVE 0.999; four key frequencies.

Seed 4. Latest grokking at step 8800; final FVE 0.997; four key frequencies; slowest circuit formation but stable final readout.

32 Progress measure definitions

Restricted (trig) loss. Mean cross-entropy on test pairs after projecting logits onto the span of $\cos/\sin(w_k(a+b-c))$ for adaptive key frequencies k.

Excluded loss. Mean cross-entropy after removing key-frequency components from logits.

Faithful trig-FVE. Fraction of logit variance explained by the key-frequency subspace, with per-(a,b) bias correction (Nanda `bias_correction=True`).

Embedding/unembedding Gini. Gini coefficient of the Fourier norm spectrum of W_E and W_U.

33 Reproducibility checklist

1. Train: `train_grokking.py --preset minimal_hiprec_act --seed $S --max-steps 50000`
2. Reproduce the paper figures: `bash scripts/reproduce_figures.sh`
3. Re-run the analysis from checkpoints: `bash scripts/run_analysis.sh`
4. Compile the manuscript: `bash compile_paper.sh`

See `REPRODUCIBILITY.md` for artifact release plan.

34 All-frequency ablation numeric table

Key frequencies from progress measures: $\{14,28,42,56,70\}$ (seed 0). Ablating any key frequency raises test CE to chance ($\approx \log 113 \approx 4.73$); ablating non-key frequencies leaves performance unchanged (Nanda Fig. 6 criterion).

Table A5: Selected per-frequency ablation losses (test CE).

Frequency k	Ablate loss	Key?
14	4.82	yes
28	4.79	yes
42	4.81	yes
7	0.02	no
21	0.02	no
35	0.02	no

35 Head trig fit methodology

We fit each head's attention from "=" to tokens a and b against $\cos/\sin(w_k \cdot \phi)$ for $\phi \in \{a, b, a+b, a-b\}$ (`head_trig_fit.py`). TRM minimal heads show $< 1\%$ FVE (Table 3): routing is near-uniform and the Fourier circuit is concentrated in SwiGLU/W_L rather than attention polynomials as in Nanda's one-layer ReLU model.

36 Figure index

Table A6 lists every figure file, its in-text label, and the corresponding figure in Nanda et al. [2023].

Table A6: Figure manifest (all under `figures/`).

File	Label	Nanda analogue
fig_algorithm_schematic.pdf	Fig. 2	Fig. 1
fig_grokking_curves.pdf	Fig. 8	Fig. 2
fig_embedding_fourier.pdf	Fig. 3	Fig. 3
fig_wl_fourier.pdf	Fig. 3	Fig. 3
fig_activations_periodic.pdf	Fig. 4	Fig. 4
fig_neuron_trig_fit.pdf	Fig. 5	Fig. 5
fig_frequency_ablation.pdf	Fig. 6	Fig. 6
fig_frequency_ablation_grid.pdf	Fig. 7	Fig. 6 (grid)
fig_progress_measures.pdf	Fig. 9	Fig. 7
fig_multiseed_summary.pdf	Fig. 10	Fig. 18
fig_fve_vs_k.pdf	Fig. 1	–
fig_data_fraction_sweep.pdf	Fig. A1	Fig. 20
fig_weight_decay_sweep.pdf	Fig. A2	Fig. 27

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